**i. Ephemerals:** These are also called **drought escapers** or **drought evaders**. These plants complete their life cycle within a short period (**single season**). These are not true xerophytes. Examples: *Argemone, Mollugo, Tribulus* and *Tephrosia*.

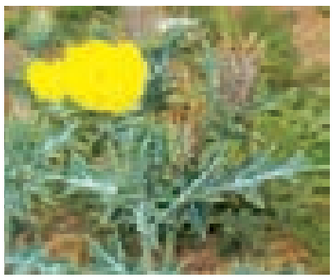


**Figure 6.20:**
*Argemone mexicana*-Ephemerals

**ii. Succulents:** These are also called drought **enduring plants**. These plants store water in their plant parts during the dry period. These plants develop certain adaptive characters to resist extreme drought conditions. Examples: *Opuntia, Aloe, Bryophyllum* and *Begonia*.

**iii. Non succulents:** These are also called **drought resistant plants ( true xerophytes).** They face both external and internal dryness. They have many adaptations to resist dry conditions. Examples: *Casuarina, Nerium, Zizyphus* and *Acacia.*

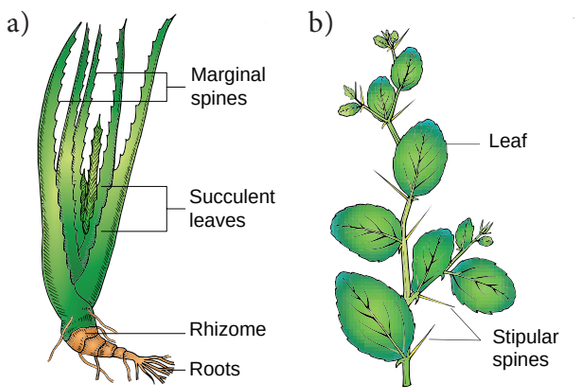



**Figure 6.21:** a)Succulent xerophyte – *Aloe*
b) Non succulent perennial - *Ziziphus*

## Morphological Adaptations

### In root

- Root system is well developed and is greater than that of shoot system.
- Root hairs and root caps are also well developed.

In xerophytic plants with the leaves and stem are covered with hairs are called **trichophyllous plants** . Example: Cucurbits (*Melothria and Mukia* )

### In stem

- Stems are mostly hard and woody. They may be aerial or underground
- The stems and leaves are covered with wax coating or covered with dense hairs.
- In some xerophytes all the internodes in the stem are modified into a fleshy leaf structure called **phylloclades** (*Opuntia*) .
- In some of the others single or occasionally two internodes modified into fleshy green structure called **cladode** (*Asparagus*).

  In some the petiole is modified into a fleshy leaf like structure called **phyllode** (*Acacia melanoxylon*).

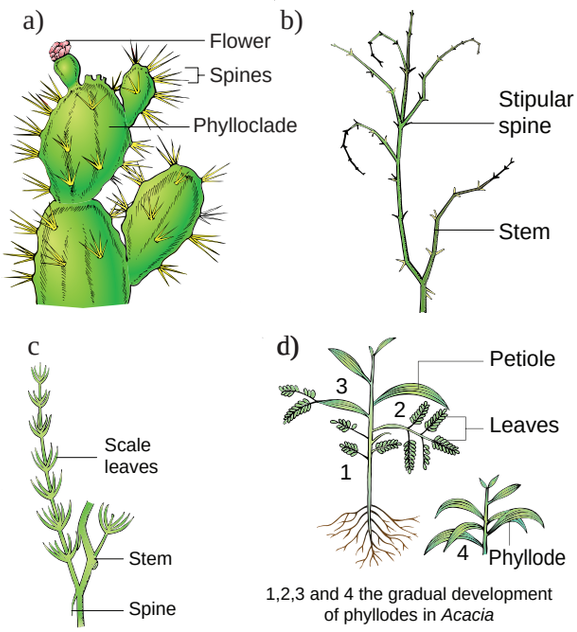



**Figure 6.22:** Xerophytes

a) A succulent xerophyte: Phylloclade – *opuntia*
b) Non succulent: Perennial - *Capparis*
c) Cladode of *Asparagus*
d) Phyllode – *Acacia*



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